

Cookies and CSRF

CS 161 Spring 2026 - Lecture 17

Announcements

- HW 4 has been released and is due this **Friday, March 6** at 11:59 PM PT.
- Project 2
  - Week 2 Design Review Sign-ups have been released!
- Midterm is coming up **(March 12th 7-9pm)**
  - The exam alterations form has been released and was due **February 20th**
  - Please submit this asap if you have yet to do so

Last Time: URLs

- URL: A string that uniquely identifies one piece of data on the web
- Parts of a URL:
  - Protocol: Defines which Internet protocol to use to retrieve the data (e.g. HTTP or HTTPS)
  - Location: Defines which web server to contact
    - Can optionally contain a username or port
  - Path: Defines which file on the web server to fetch
  - Query (optional): Sends arguments in name-value pairs to the web server
  - Fragment (optional): Not sent to the web server, but used by the browser for processing
- Special characters should be URL escaped

Last Time: HTTP

- HTTP: A protocol used to request and retrieve data from a web server
  - HTTPS: A secure version of HTTP
  - HTTP is a request-response protocol
- HTTP request
  - Method (GET or POST)
  - URL path and query parameters
  - Protocol
  - Data (only for POST requests)
- HTTP response
  - Protocol
  - Status code: A number indicating what happened with the request
  - Headers: Metadata about the response
  - Data

Last Time: Parts of a Webpage

- HTML: A markup language to create structured documents
  - Create a link
  - Create a form
  - Embed an image
  - Embed another webpage (iframe or frame)
- CSS: A style sheet language for defining the appearance of webpages
  - As powerful as JavaScript if used maliciously!
- JavaScript: A programming language for running code in the web browser
  - JavaScript code runs in the web browser
  - Modify any part of the webpage (e.g. HTML or CSS)
  - Create pop-up messages
  - Make HTTP requests

Today: Mysteries

- How is it that I can open Gmail today and read my email, even though I didn't log in today? How does it remember me?
- How is it that if I'm logged out from Google, and then I log into my Google account in one browser window, my Google docs in other browser windows update to show me as logged in, even though I didn't log in in those windows?

Today: Cookies and CSRF

- Cookies
  - Parts of a cookie
- Cookie Policy
  - Setting cookies
  - Sending cookies
- Session Authentication
- Cross-Site Request Forgery (CSRF)
- CSRF Defenses





Cookies

Customizing HTTP Responses

- HTTP is a request-response protocol
  - The web server processes each request independently of other requests
- What if we want our responses to be customized?
  - Example: If I enable dark mode on a website, I want future responses from the website to be in dark mode
  - Example: If I log in to a website, I want future responses from the website to be related to my account
- Cookies are a way to add state to HTTP

Cookies: Definition

- **Cookie**: a piece of data used to maintain state across multiple requests
- This data is stored and remembered by the browser, and sent to server when needed
- Creating cookies
  - The server can create a cookie by including a `Set-Cookie` header in its response
  - JavaScript running in the browser can create a cookie
- Storing cookies
  - Cookies are stored by the web browser (not the web server)
  - The browser's cookie storage is sometimes called a **cookie jar**
- Sending cookies
  - The browser *automatically* attaches relevant cookies to every request
  - The server uses received cookies to customize responses and connect related requests

Parts of a Cookie: Name and Value

- The actual data in the cookie is stored as a **name-value pair**

Name	`Theme`
Value	`Dark`
Domain	`toon.cs161.org`
Path	`/themes`
Secure	`True`
HttpOnly	`False`
Expires	`11 Jul 2024 20:00:00`
(*other fields omitted*)	

Parts of a Cookie: Domain and Path

- The **domain attribute** and **path attribute** define which requests the browser should attach this cookie for
- The domain attribute usually looks like the domain in a URL
- The path attribute usually looks like a path in a URL

Name	`Theme`
Value	`Dark`
Domain	`toon.cs161.org`
Path	`/themes`
Secure	`True`
HttpOnly	`False`
Expires	`11 Jul 2024 20:00:00`
(*other fields omitted*)	

Parts of a Cookie: Secure and HttpOnly

- The Secure attribute and HttpOnly attribute restrict the cookie for security purposes
- If the **Secure attribute** is True, then the browser only sends the cookie if the request is made over HTTPS (and doesn't send the cookie with HTTP requests)
- If the **HttpOnly attribute** is True, then JavaScript in the browser is not allowed to access the cookie

Name	`Theme`
Value	`Dark`
Domain	`toon.cs161.org`
Path	`/themes`
Secure	`True`
HttpOnly	`False`
Expires	`11 Jul 2024 20:00:00`
(*other fields omitted*)	

Parts of a Cookie: Expires

- The **Expires attribute** defines when the cookie is no longer valid
- The expires attribute is usually a timestamp
- If the timestamp is in the past, then the cookie has expired, and the browser deletes it from the cookie jar

Name	`Theme`
Value	`Dark`
Domain	`toon.cs161.org`
Path	`/themes`
Secure	`True`
HttpOnly	`False`
Expires	`11 Jul 2024 20:00:00`
(*other fields omitted*)	

Cookie Policy

Cookies: Issues

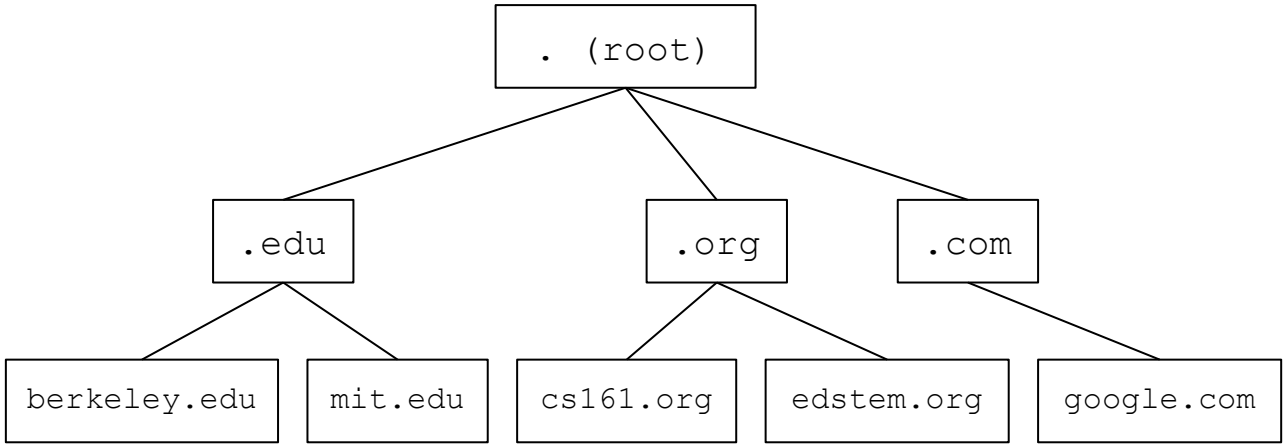
- Recall:
  - The server can create a cookie by including a **`Set-Cookie`** header in its response
  - The browser automatically attaches relevant cookies in every request
- Security issues:
  - A server should not be able to set cookies for unrelated websites
    - Example: **`evil.com`** should not be able to set a cookie that gets sent to **`google.com`**
  - Cookies shouldn't be sent to the wrong websites
    - Example: A cookie used for authenticating a user to Google should not be sent to **`evil.com`**
    - We'll see how cookies are used for logins later

Cookie Policy

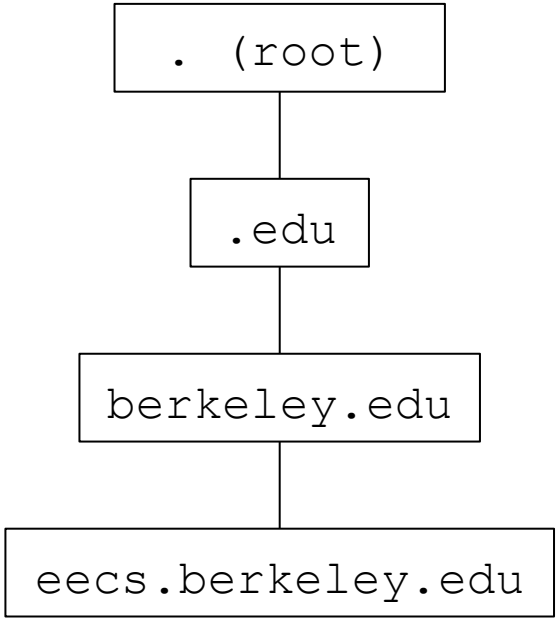
- **Cookie policy**: A set of rules enforced by the browser
  - When the browser receives a cookie from a server, should the cookie be accepted?
  - When the browser makes a request to a server, should the cookie be attached?
- Cookie policy is (confusingly) different from the same-origin policy

Domain Hierarchy

- Recall: Domains
  - Located after the double slashes, but before the next single slash
  - Written as several phrases separated by dots
- Domains can be sorted into a hierarchy
  - The hierarchy is separated by dots



Domain Hierarchy



`.edu` is a **top-level domain** (TLD), because it is directly below the root of the tree.

`eecs.berkeley.edu` is a **subdomain** of `berkeley.edu`.

Cookie Policy 1: Setting Cookies

- When the browser receives a cookie from a server, should the cookie be accepted?
- Server with domain X can set a cookie whose domain attribute is Y if
  - (1) The domain attribute is a **domain suffix** of the server's domain
    - X ends in Y
    - X is below or equal to Y on the hierarchy
    - X is more specific or equal to Y
  - ... and, (2) the domain attribute Y is not a top-level domain (TLD)
  - No restrictions for the Path attribute (the browser will accept any value)
- Examples:
  - mail.**google.com** can set cookies for Domain=mail.google.com or Domain=google.com
  - **google.com** can set cookies for Domain=google.com
  - google.**com** **cannot** set cookies for Domain=com, because com is a top-level domain

Cookie Policy 2: Sending Cookies

- When the browser makes a request to a server, should the cookie be attached?
- The browser sends the cookie if both of these are true:
  - The domain attribute is a **domain suffix** of the server's domain
  - The path attribute is a **prefix** of the server's path

Intuition: Domain attributes

- If the domain attribute is set to `google.com`, think of this as representing `google.com` or `*.google.com` or `*.*.google.com` or ...
- A server can only set a cookie if the server matches the cookie's domain attribute
- A cookie is only sent to a server if the server matches the domain attribute

Cookie Policy: Sending Cookies

(server URL)

`https://toon.cs161.org/cryptoverse/oneshots/subway.html`

`cs161.org/cryptoverse`

(cookie domain) (cookie path)

Quick method to check cookie sending:
- Right-align the domains. The cookie domain should be a suffix of the server URL.

Cookie Policy: Sending Cookies

(server URL)

`https://toon.cs161.org/cryptoverse/oneshots/subway.html`

`cs161.org/cryptoverse`

(cookie domain) (cookie path)

Quick method to check cookie sending:

- Right-align the domains. The cookie domain should be a suffix of the server URL.
- Left-align the paths. The cookie domain should be a prefix of the server path.

Cookie Policy: Sending Cookies

(server URL)

`https://toon.cs161.org/cryptoverse/oneshots/subway.html`

`cs161.org/cryptoverse`

(cookie domain) (cookie path)

Quick method to check cookie sending: Concatenate the cookie domain and path. Line it up below the requested URL at the first single slash.

If the domains and paths all match, then the cookie is sent.

Cookie Policy: Sending Cookies

(server URL)

`https://toon.cs161.org/cryptoverse/oneshots/subway.html`

`cs161.org/themes`

(cookie domain) (cookie path)

Quick method to check cookie sending: Concatenate the cookie domain and path. Line it up below the requested URL at the first single slash.

If the domain or path doesn't match, then the cookie is not sent.

Cookie Policy: Sending Cookies

(server URL)

`https://su24.cs161.org/proj2/story`

`www.cs161.org/proj2`

(cookie domain) (cookie path)

Will a cookie with this domain and path be sent to this server?

Cookie Policy: Sending Cookies

(server URL)

`https://su24.cs161.org/proj2/story`

`www.cs161.org/proj2`

(cookie domain) (cookie path)

Will a cookie with this domain and path be sent to this server?

No! The domain does not match!

Examples: Which cookie will be sent?

cookie 1	cookie 2	cookie 3
name = id value = chocolate domain = login.site.com path = / non-secure	name = id value = vanilla domain = .site.com path = / non-secure	name = userid value = plain domain = login.site.com path = / secure

http://checkout.site.com/ — **cookie: id=vanilla**

http://login.site.com/ — **cookie: id=chocolate, id=vanilla**

http://othersite.com/ — **cookie: none**

https://login.site.com/ — **cookie: id=plain; id=vanilla; id=chocolate**

Session Authentication

Session Authentication

- **Session**: A sequence of requests and responses associated with the same authenticated user
  - Example: When you check all your unread emails, you make many requests to Gmail. The Gmail server needs a way to know all these requests are from you
  - When the session is over (you log out, or the session expires), future requests are not associated with you
- Naïve solution: Type your username and password before each request
  - Problem: Very inconvenient for the user!
- Better solution: Is there a way the browser can automatically send some information in a request for us?
  - Yes: Cookies!

Session Authentication: Intuition

- Imagine you're attending a concert
- The first time you enter the venue:
  - Present your ticket and ID
  - The doorperson checks your ticket and ID
  - If they're valid, you receive a wristband
- If you leave and want to re-enter later
  - Just show your wristband!
  - No need to present your ticket and ID again



Session Tokens

- **Session token**: A secret value used to associate requests with an authenticated user
- The first time you visit the website:
  - Present your username and password
  - If they're valid, you receive a session token
  - The server associates you with the session token
- When you make future requests to the website:
  - Attach the session token in your request
  - The server checks the session token to figure out that the request is from you
  - No need to re-enter your username and password!

Session Tokens with Cookies

- Session tokens can be implemented with cookies
  - Cookies can be used to save *any* state across requests (e.g., dark mode)
  - Session tokens are just one (common) way to use cookies
- The first time you visit a website:
  - Make a request with your username and password
  - If they're valid, the server sends you a cookie with the session token
  - The server associates you with the session token
- When you make future requests to the website:
  - The browser attaches the session token cookie in your request
  - The server checks the session token to figure out that the request is from you
  - No need to re-enter your username and password!
- When you log out (or when the session times out):
  - The browser and server delete the session token

The Basic Idea: Intuition (A Close Approximation)

Login/session initiation:

- Server sends the browser a cookie. The cookie contains a random, unique value that identifies the user: e.g., *sessionid=09c3ad..9f*

  (The cookie is attached to every subsequent request to the same server that originally set the cookie)

When the user visits the same website:

- The browser sends this value along with every request to that domain
- The server remembers the association sessionid=09c3ad..9f ⇄ userid=ralucapopa
- This way, the server can associate each request to the user who is logged in

Session Tokens: Security

- If an attacker steals your session token, they can log in as you!
  - The attacker can make requests and attach your session token
  - The server will think the attacker's requests come from you
- Servers need to generate session tokens *randomly* and *securely*
- Browsers need to make sure malicious websites cannot steal session tokens
  - Enforce isolation with cookie policy and same-origin policy
- Browsers should not send session tokens to the wrong websites
  - Enforced by cookie policy

Session Token Cookie Attributes

- What attributes should the server set for the session token?

Session Token Cookie Attributes

- What attributes should the server set for the session token?
  - Domain and Path: Set so that the cookie is only sent on requests that require authentication
  - Secure: Set to True to so the cookie is only sent over secure HTTPS connections
  - HttpOnly: Set to True so JavaScript can't access session tokens
  - Expires: Set so that the cookie expires when the session times out

Name	`token`
Value	`{random value}`
Domain	`mail.google.com`
Path	`/`
Secure	`True`
HttpOnly	`True`
Expires	`{15 minutes later}`
(*other fields omitted*)	

Cross-Site Request Forgery (CSRF)



Review: Cookies and Session Tokens

- Session token cookies are used to associate a request with a user
- The browser automatically attaches relevant cookies in every request

Cross-Site Request Forgery (CSRF)

- Idea: What if the attacker tricks the victim into making an unintended request?
  - The victim's browser will automatically attach relevant cookies
  - The server will think the request came from the victim!
- **Cross-site request forgery (CSRF or XSRF)**: An attack that exploits cookie-based authentication to perform an action as the victim

Analogy: Passing Notes to My Banker

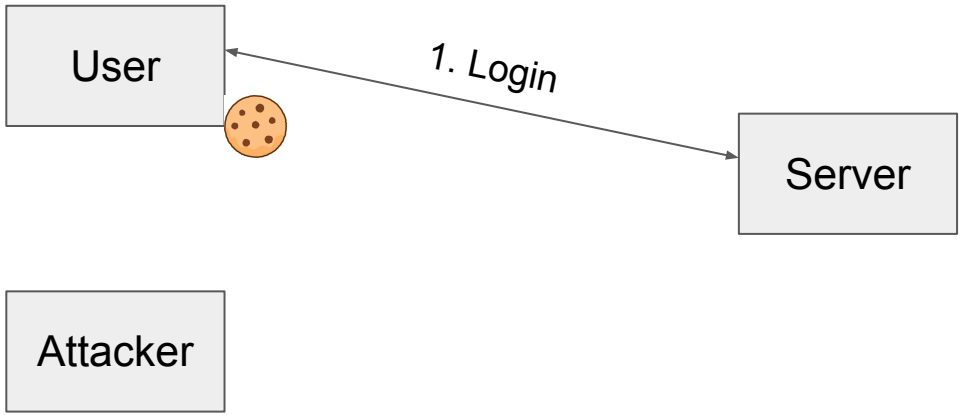
- I have a bank account with a local bank, with $1000 in my checking account.
- I can pay someone at any time by writing out a check (a note that is verified to come from me that says, e.g., "Please pay $50 to Alice from my account."), and giving this check to my bank
- I'm at a party, and a stranger Genevieve confides to me that she has a crush on a guy there and convinces me to pass a romantic note to him without reading it.
- Unbeknownst to me, the note actually says "Please pay $1000 to Genevieve from my account" ... and the guy is my banker. When he receives the note, he assumes I want to pay Genevieve and transfers money from my account to her.
- The flaw: My banker assumed that if a request came from me, I must have authorized it.

Steps of a CSRF Attack



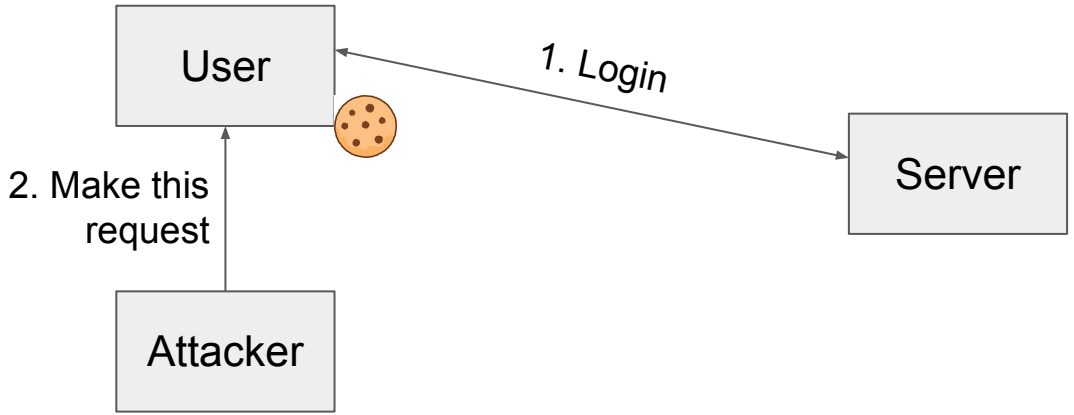
Steps of a CSRF Attack

1. User authenticates to the server
    - User receives a cookie with a valid session token



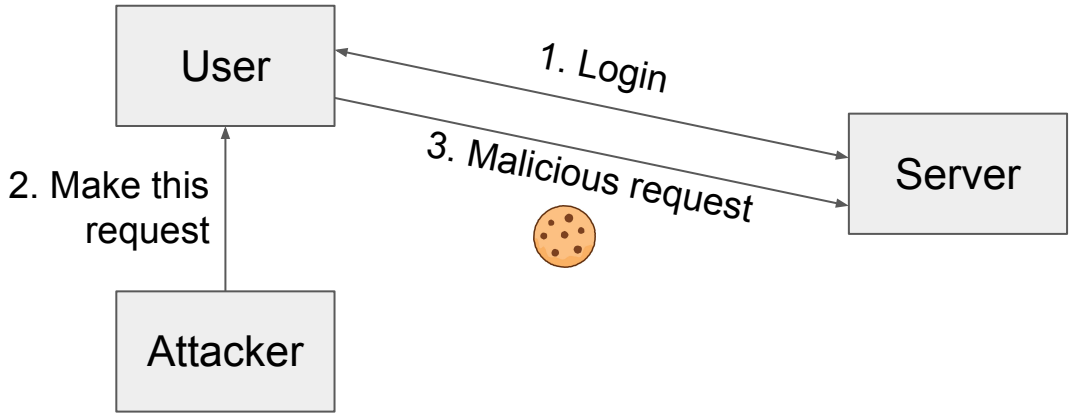
Steps of a CSRF Attack

1. User authenticates to the server
   - User receives a cookie with a valid session token
2. Attacker tricks the victim into making a malicious request to the server



Steps of a CSRF Attack

1. User authenticates to the server
   - User receives a cookie with a valid session token
2. Attacker tricks the victim into making a malicious request to the server
3. The server accepts the malicious request from the victim
   - Recall: The cookie is automatically attached in the request



Steps of a CSRF Attack

1. User authenticates to the server
    - User receives a cookie with a valid session token
2. Attacker tricks the victim into making a malicious request to the server
3. The server accepts the malicious request from the victim
    - Recall: The cookie is automatically attached in the request

Executing a CSRF Attack

- How might we trick the victim into making a GET request?

Executing a CSRF Attack

- How might we trick the victim into making a GET request?
- Strategy #1: Trick the victim into clicking a link
  - For example, send the victim an email with an enticing-looking link
  - The link can directly make a GET request:

    `https://www.bank.com/transfer?amount=100&to=Mallory`
  - The link can open an attacker's website, which contains some JavaScript that makes the actual malicious request
- Strategy #2: Put some HTML on a website the victim will visit
  - Example: The victim will visit a forum. Make a post with some HTML on the forum
  - HTML to automatically make a GET request to a URL:

    `<img src="https://www.bank.com/transfer?amount=100&to=Mallory">`
    - This HTML will probably return an error or a blank 1 pixel by 1 pixel image, but the GET request will still be sent...with the relevant cookies!

Executing a CSRF Attack

- How might we trick the victim into making a POST request?
  - Example POST request: Submitting a form
- Strategy #1: Trick the victim into clicking a link
  - Note: Clicking a link in your browser makes a GET request, not a POST request, so the link cannot directly make the malicious POST request
  - The link can open an attacker's website, which contains some JavaScript that makes the actual malicious POST request
- Strategy #2: Put some JavaScript on a website the victim will visit
  - Example: Pay for an advertisement on the website, and put JavaScript in the ad
  - Recall: JavaScript can make a POST request

Top 25 Most Dangerous Software Weaknesses (2020)

Rank	ID	Name	Score
[1]	CWE-79	Improper Neutralization of Input During Web Page Generation ('Cross-site Scripting')	46.82
[2]	CWE-787	Out-of-bounds Write	46.17
[3]	CWE-20	Improper Input Validation	33.47
[4]	CWE-125	Out-of-bounds Read	26.50
[5]	CWE-119	Improper Restriction of Operations within the Bounds of a Memory Buffer	23.73
[6]	CWE-89	Improper Neutralization of Special Elements used in an SQL Command ('SQL Injection')	20.69
[7]	CWE-200	Exposure of Sensitive Information to an Unauthorized Actor	19.16
[8]	CWE-416	Use After Free	18.87
[9]	CWE-352	Cross-Site Request Forgery (CSRF)	17.29
[10]	CWE-78	Improper Neutralization of Special Elements used in an OS Command ('OS Command Injection')	16.44
[11]	CWE-190	Integer Overflow or Wraparound	15.81
[12]	CWE-22	Improper Limitation of a Pathname to a Restricted Directory ('Path Traversal')	13.67
[13]	CWE-476	NULL Pointer Dereference	8.35
[14]	CWE-287	Improper Authentication	8.17
[15]	CWE-434	Unrestricted Upload of File with Dangerous Type	7.38
[16]	CWE-732	Incorrect Permission Assignment for Critical Resource	6.95
[17]	CWE-94	Improper Control of Generation of Code ('Code Injection')	6.53

CSRF Example: Internet of Things (IoT)

- IoT devices often use default passwords
  - Example login URL: `http://10.1.1.1/login?user=admin&password=admin`
  - The attacker tricks the victim into making a request to this link, which gives the victim a session token cookie
- IoT devices often process commands with web requests
  - Example request URL: `http://10.1.1.1/set-dns-server?server=8.8.8.8`
  - These requests can perform actions that help the attacker mount other attacks
- IoT devices don't implement CSRF defenses
  - An attacker can publish an advertisement that makes these two requests
  - Any victim who opens a webpage with the advertisement will get their IoT device compromised!

CSRF Example: Malvertising

- Short for **malicious advertising**
- The attacker writes some JavaScript
  - The script opens a 1 pixel by 1 pixel frame (too small for human users to notice)
  - The frame opens a huge number of internal frames
  - Each frame launches possible CSRF attacks
- The attacker pays for advertisements to put this JavaScript on legitimate websites
- **Takeaway**: Advertisers might publish malicious scripts for you as long as you pay for it

CSRF Example: YouTube

- 2008: Attackers exploit a CSRF vulnerability on YouTube
- By forcing the victim to make a request, the attacker could:
  - Make the victim share a video with their contacts
  - Make the victim subscribe to any channel
  - Add any videos to the user's watchlist
- **Takeaway**: With a CSRF attack, the attacker can force the victim to perform a wide variety of actions!

CSRF Example: Facebook

InternetNews.com

Link

Facebook Hit by Cross-Site Request Forgery Attack

Sean Michael Kerner *August 21, 2009*

Nevertheless, that Facebook accounts were compromised in the wild is noteworthy because the attack used a legitimate HTML tag to violate users' privacy.

According to Zilberman's disclosure, the attack simply involved the malicious HTML image tag residing on any site, including any blog or forum that permits the use of image tags even in the comments section.

"The attack elegantly ends with a valid image so the page renders normally, and the attacked user does not notice that anything peculiar has happened," Zilberman said.

Takeaway: The HTML image tag can be used to execute a CSRF attack

Cookies: Summary

- Cookie: a piece of data used to maintain state across multiple requests
  - Set by the browser or server
  - Stored by the browser
  - Attributes: Name, value, domain, path, secure, HttpOnly, expires
- Cookie policy
  - Server with domain X can set a cookie with domain attribute Y if the domain attribute is a **domain suffix** of the server's domain, and the domain attribute Y is not a top-level domain (TLD)
  - The browser attaches a cookie on a request if the domain attribute is a **domain suffix** of the server's domain, and the path attribute is a **prefix** of the server's path

Session Authentication: Summary

- Session authentication
  - Use cookies to associate requests with an authenticated user
  - First request: Enter username and password, receive session token (as a cookie)
  - Future requests: Browser automatically attaches the session token cookie
- Session tokens
  - If an attacker steals your session token, they can log in as you
  - Should be randomly and securely generated by the server
  - The browser should not send tokens to the wrong place

CSRF: Summary

- Cross-site request forgery (CSRF or XSRF): An attack that exploits cookie-based authentication to perform an action as the victim
  - User authenticates to the server
    - User receives a cookie with a valid session token
  - Attacker tricks the victim into making a malicious request to the server
  - The server accepts the malicious request from the victim
    - Recall: The cookie is automatically attached in the request
- Attacker must trick the victim into creating a request
  - GET request: click on a link
  - POST request: use JavaScript